Other Technologies Related to Software Development

BIS605

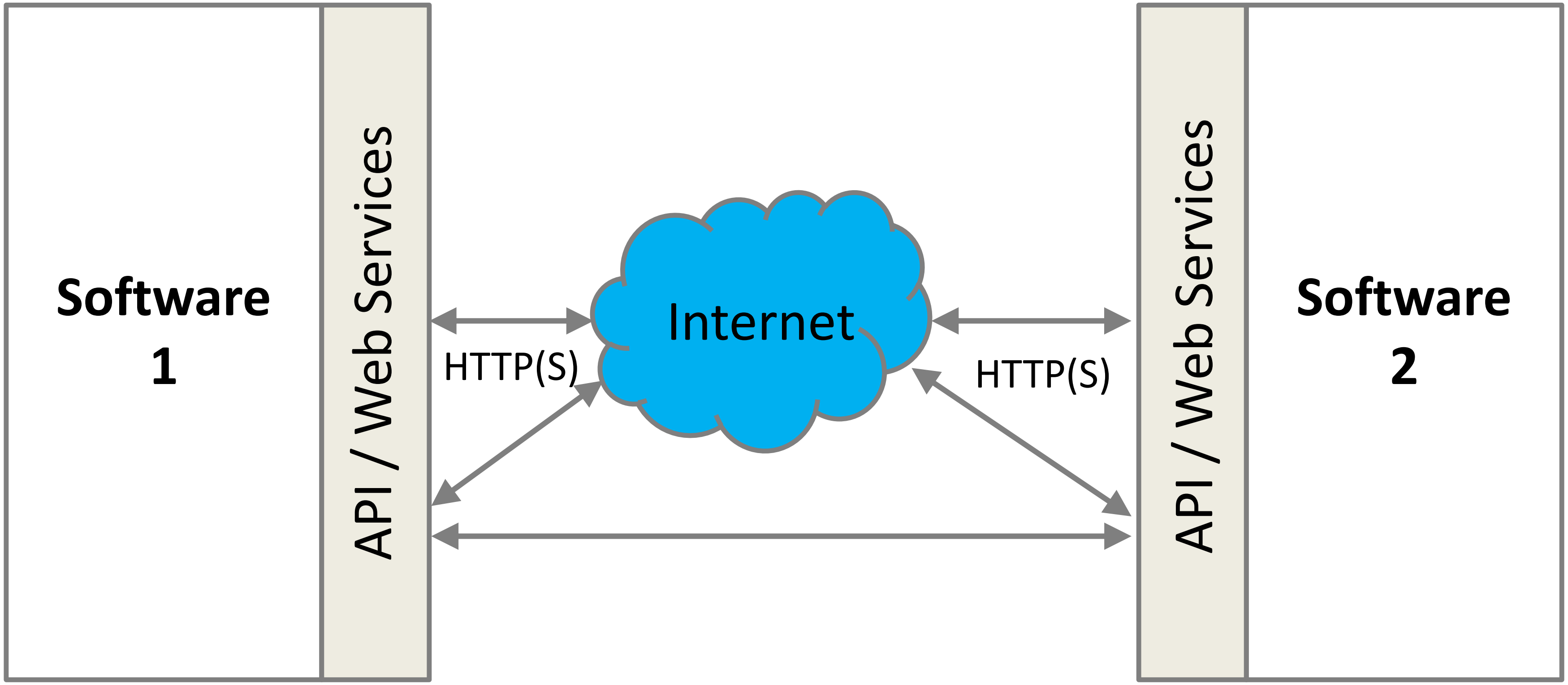
Software Development Technologies for Digital Business

Asst. Prof. Dr. Worarat Krathu

Resources

- William EL Kaim, Introduction to API, at slideshare.net
- Krish, Web Services, at slideshare.net
- Gerard Sylvester, Webservices, at slideshare.net
- Kabir Baidya, RESTful Architecture, at slideshare.net
- Simplilearn, What is Cloud Computing?, at slideshare.net
- https://phppot.com/php/php-restful-web-service
- https://en.wikipedia.org/wiki/Web_Services_Description_Language
- https://nordicapis.com/5-examples-of-apis-we-use-in-our-everyday-lives/
- https://en.wikipedia.org/wiki/Mobile_app_development
- https://en.wikipedia.org/wiki/Mobile_app
- https://www.outsource2india.com/software/mobile-applications/tools-technologies.asp
- https://careerfoundry.com/en/blog/web-development/what-is-the-difference-between-a-mobile-app-and-a-web-app/
- Introduction to the Internet of Things, Lecture of INT690

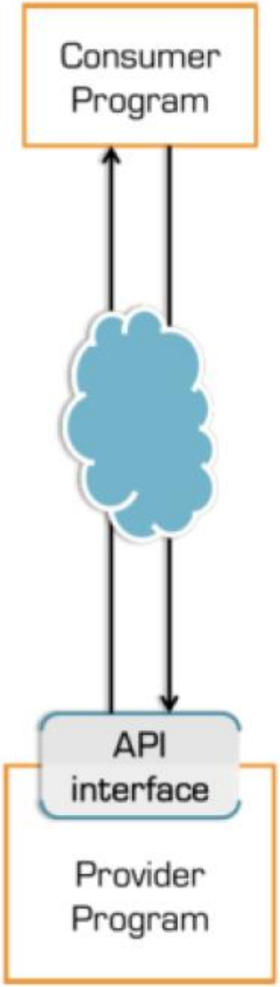
Software and Related Components



API AND WEB SERVICES

API

- An Application Programming Interface (API) is a specification intended to be used as an interface by software components to communicate with each other

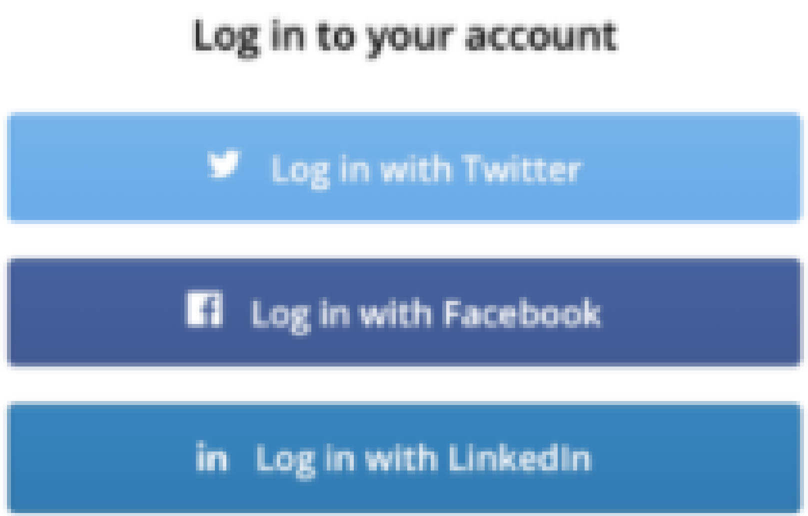


API

- Example – Facebook Like

API

- Example – Log-in
  - Facebook
  - Linkedin
  - Twitter
  - Google



API Typology

- Private – internally to facilitate the integration of different applications and systems used by a company
- Partner – facilitate communication and integration of software between a company and its business partners
- Public – publicly expose information and functionalities of one or various systems and applications to third parties that do not necessarily have a business relationship with them

Web Service

- Web service is a network accessible interface to application functionality, built using standard Internet protocols

- It exposes functionality to consumer
  - Over the Internet of intranet
  - A programmable URL
  - Functions can be called over the Internet

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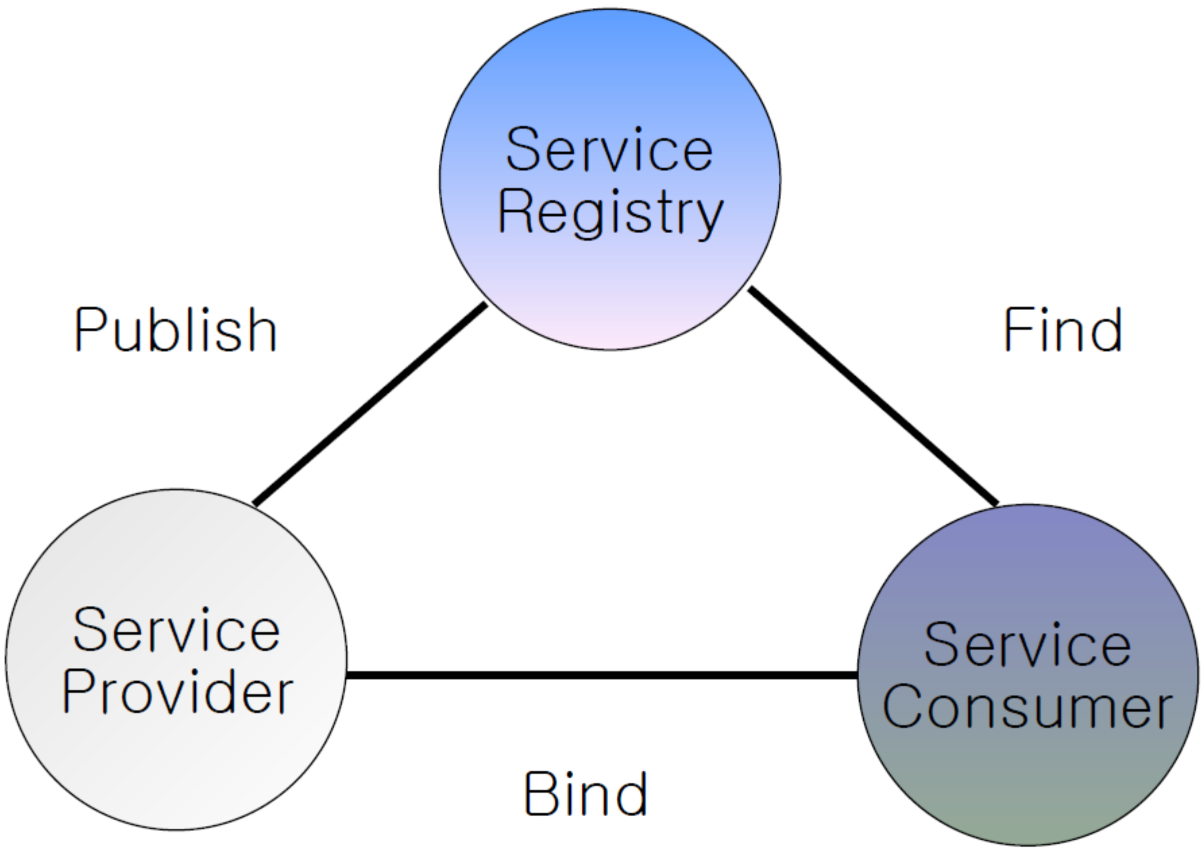
Web Service Architecture

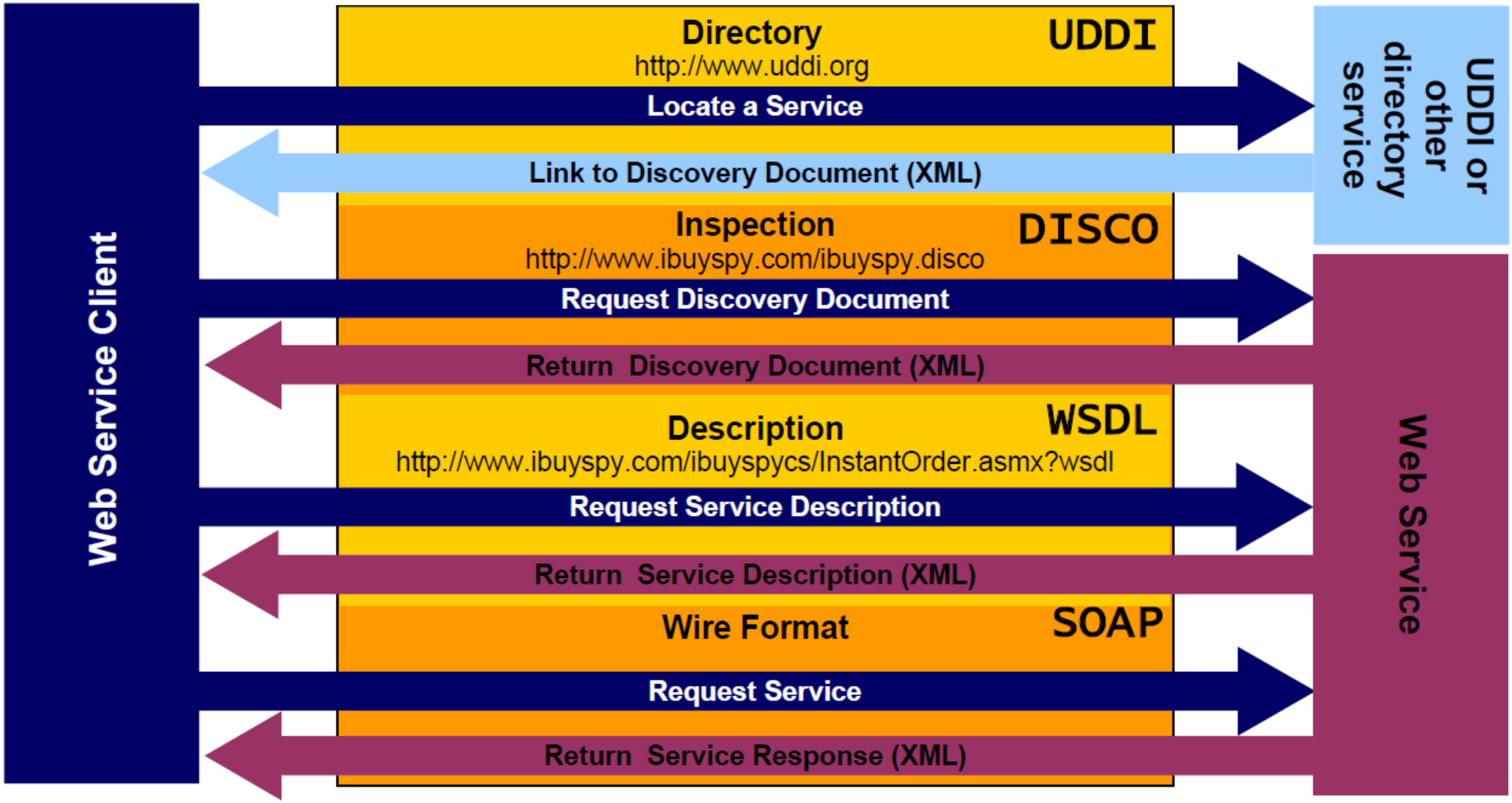


Web Service with SOAP and WSDL

- A standard way of communication (SOAP)
- A uniform data representation and exchange mechanism (XML)
- A standard meta language to describe the services offered (WSDL)
- A mechanism to register and locate WS based applications (UDDI)

Directory: Publish & Find Services:	UDDI
Inspection: Find Services on server:	DISCO
Description: Formal Service Descriptions:	WSDL
Wire Format: Service Interactions:	SOAP
Universal Data Format:	XML
Ubiquitous Communications:	Internet

Components of a Web Service



Web Service - UDDI

- Universal Description, Discovery and Integration (UDDI) protocol
- Enables quickly and dynamically discover and invoke Web Services both internally and externally
- Yellow pages of Web Services
- Examples
  - www.uddi.org
  - www.biomoby.org
  - www.xmethods.com

Web Service - WSDL

- Web Service Description Language
- Describing the functionality offered by a Web Service

```xml
<message name="getTermRequest">
  <part name="term" type="xs:string"/>
</message>

<message name="getTermResponse">
  <part name="value" type="xs:string"/>
</message>

<portType name="glossaryTerms">
  <operation name="getTerm">
    <input message="getTermRequest"/>
    <output message="getTermResponse"/>
  </operation>
</portType>

<binding type="glossaryTerms" name="b1">
   <soap:binding style="document"
   transport="http://schemas.xmlsoap.org/soap/http" />
   <operation>
     <soap:operation soapAction="http://example.com/getTerm"/>
     <input><soap:body use="literal"/></input>
     <output><soap:body use="literal"/></output>
  </operation>
</binding>
```

Source: https://www.w3schools.com/xml/xml_wsdl.asp

Web Service - SOAP

- Simple Object Access Protocol (SOAP)
- A lightweight (XML-based) protocol for exchange of information in decentralized, distributed environment
  - An envelope that defines a framework for describing what is in a message and how to process it
  - Relies heavily on XML standards (schemas & name spaces)

```xml
<?xml version="1.0"?>

<soap:Envelope
xmlns:soap="http://www.w3.org/2003/05/soap-envelope/"
soap:encodingStyle="http://www.w3.org/2003/05/soap-encoding">

<soap:Body xmlns:m="http://www.example.org/stock">
  <m:GetStockPrice>
    <m:StockName>IBM</m:StockName>
  </m:GetStockPrice>
</soap:Body>

</soap:Envelope>
```

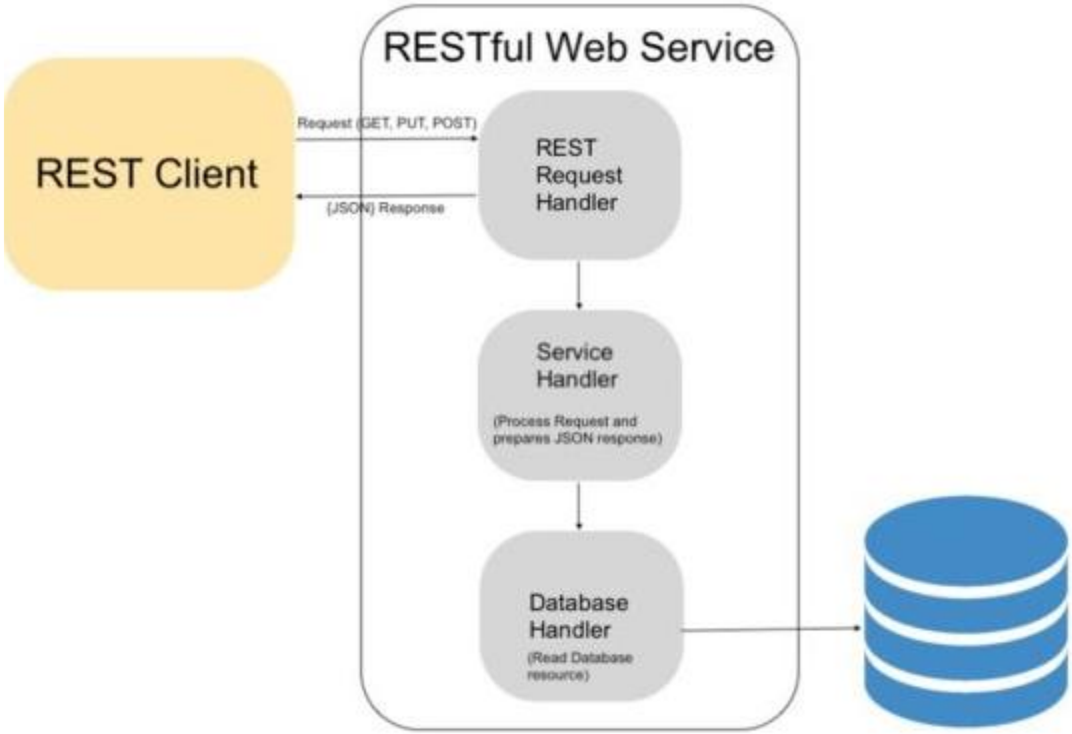
```xml
<?xml version="1.0"?>

<soap:Envelope
xmlns:soap="http://www.w3.org/2003/05/soap-envelope/"
soap:encodingStyle="http://www.w3.org/2003/05/soap-encoding">

<soap:Body xmlns:m="http://www.example.org/stock">
  <m:GetStockPriceResponse>
    <m:Price>34.5</m:Price>
  </m:GetStockPriceResponse>
</soap:Body>

</soap:Envelope>
```

Source: https://www.w3schools.com/xml/xml_soap.asp

Web Service with REST

- Representational State Transfer (REST)
- A simpler alternative to SOAP, WSDL-based Web Services
- Typically based on HTTP, but not always
- Why REST?
  - Simplicity & ease of use
  - Made for web
  - Both machine & human friendly
  - Scalability, modifiability, portability etc.

RESTful Web Services API Architecture



REST APIs

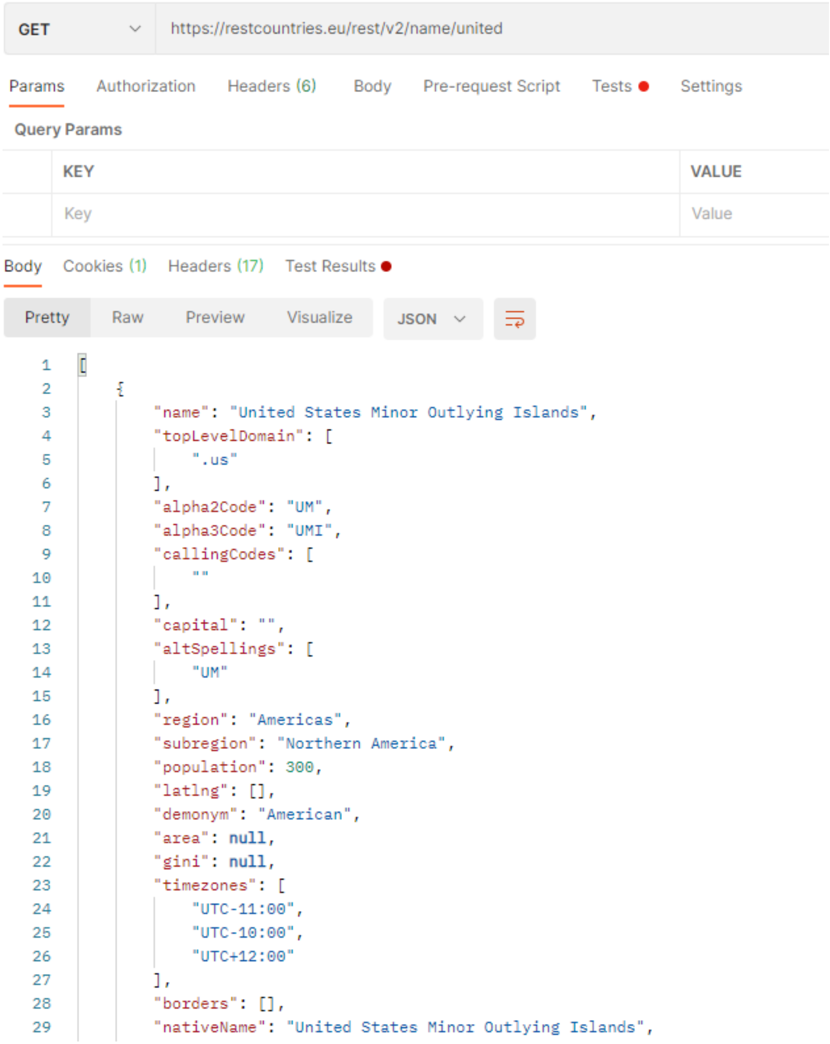
- APIs that based upon REST architecture having constraints as follows:
  - Client-sever
  - Statelessness
  - Cacheability
  - Layered system
  - Code on demand
  - Uniform interface

REST APIs

- Platform independent
  - E.g. client written in Java can communicate with APIs written in PHP, Node etc.
- Can be developed using any server-side programming languages: PHP, Ruby, Python, NodeJS, Java, C#, etc.
- User URIs to identify particular resource on server
- HTTP Verbs to represent actions (POST, GET, PUT/PATCH, DELETE)
- Response in: JSON, XML, CSV, HTML etc.

Web Service with REST

- Example
  - https://restcountries.eu/



Web Service with REST

- Example
  - https://www.omdbapi.com/

Request:

http://www.omdbapi.com/?t=hello

Response:

{"Title":"Hello","Year":"2008","Rated":"N/A","Released":"10 Oct 2008","Runtime":"129 min","Genre":"Drama, Romance","Director":"Atul Agnihotri","Writer":"Atul Agnihotri (screenplay), Chetan Bhagat (additional dialogue), Chetan Bhagat (book), Jalees Sherwani (lyrics), Alok Upadhyay (additional dialogue)","Actors":"Sharman Joshi, Amrita Arora, Sohail Khan, Isha Koppikar","Plot":"Call-center workers receive a phone call from God.","Language":"Hindi","Country":"India","Awards":"N/A","Poster":"https://m.media-amazon.com/images/M/MV5BZGM5NjliODgtODVlOS00OWZmLWIzYzMtMTI2OWIzMTM1ZGRhXkEyXkFqcGdeQXVyNDUzOTQ5MjY@._V1_SX300.jpg","Ratings":[{"Source":"Internet Movie Database","Value":"3.3/10"},{"Source":"Rotten Tomatoes","Value":"19%"}],"Metascore":"N/A","imdbRating":"3.3","imdbVotes":"1,892","imdbID":"tt1087856","Type":"movie","DVD":"05 Apr 2018","BoxOffice":"N/A","Production":"N/A","Website":"N/A","Response":"True"}

CLOUD COMPUTING

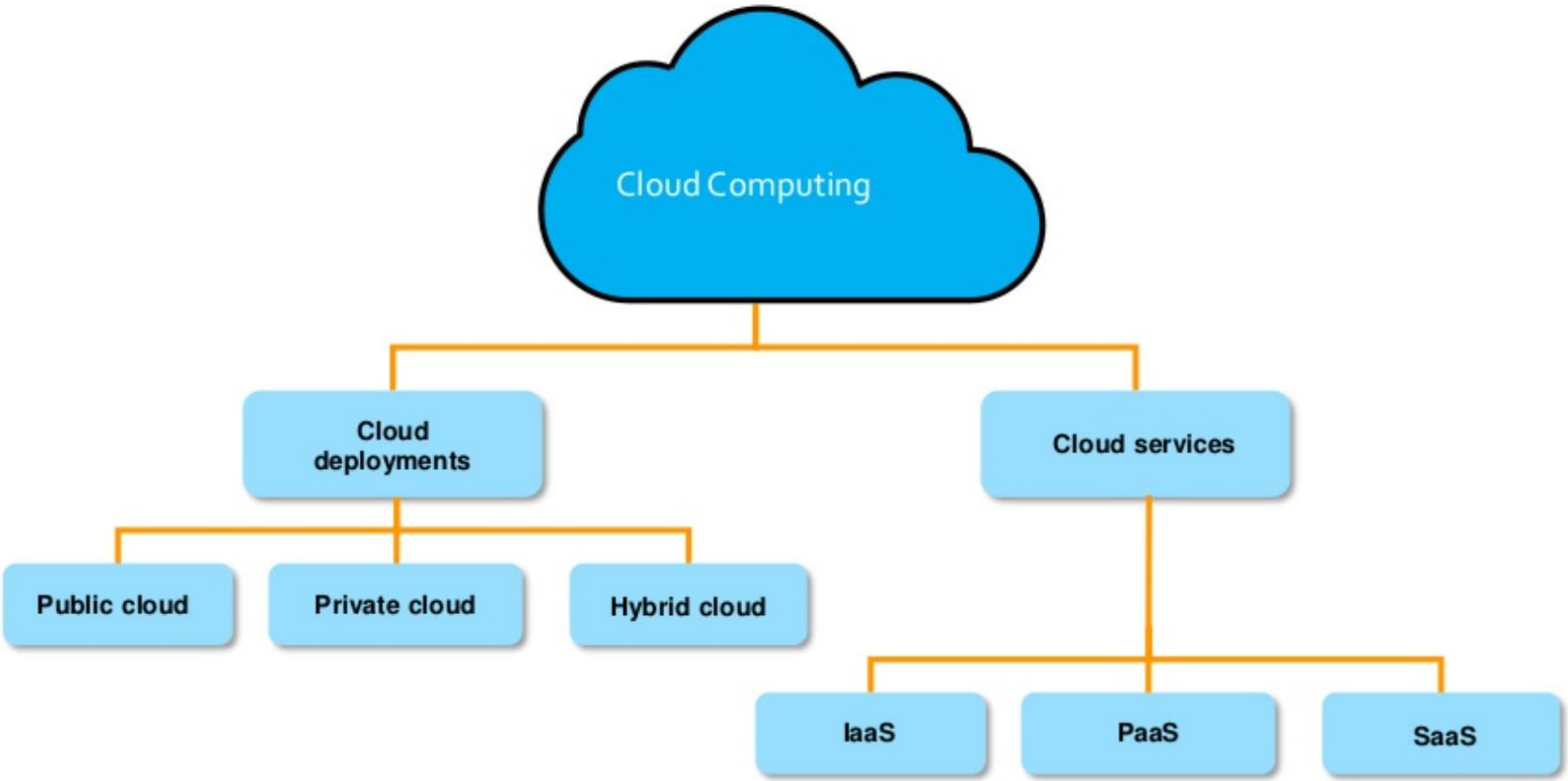
Cloud Computing

- Cloud Computing is the use of network of remote servers hosted on the Internet to store, manage and process data rather than a local server
- It delivers computing services over the internet
- Clients pay for what they use
- Cloud computing service providers give the ability to manage applications and services through a global network
- Example
  - AWS
  - Azure

Cloud Computing Benefit

- Speed - vast amount of recourses can be provisioned in minutes
- Cost – eliminates the expense of buying computer hardware and software
- Scalability – easy to scale up
- Accessibility – easy to access data anywhere
- Better security – with cloud your data is stored in a centralized secure location

Type of Cloud Computing



Deployment Model

- Public cloud
  - Services are store off-site and accessed over internet
  - All hardware software and other infrastructure is owned and managed by the provider
- Private cloud
  - Infrastructure is used exclusively by a single organization
  - Organization may run its private cloud or outsource it to a hosting company
  - Services and infrastructure are maintained on a private network
- Hybrid cloud
  - Consists the functionalities of both public and private cloud

Service Model

- Infrastructure as a service (IaaS)
  - IT infrastructure is provided from a cloud provider (AWS Elastic Compute Cloud)
- Platform as a service (PaaS)
  - Provides platform on which software can be developed and deployed (AWS Elastic Beanstalk)
- Software as a service (SaaS)
  - Cloud provider host and manage the software application on a subscription basis (Amazon Web Services)

Service Model

On-Premises	IaaS	PaaS	SaaS
Applications	Applications	Applications	Applications
Data	Data	Data	Data
Runtime	Runtime	Runtime	Runtime
Middleware	Middleware	Middleware	Middleware
O/S	O/S	O/S	O/S
Virtualization	Virtualization	Virtualization	Virtualization
Servers	Servers	Servers	Servers
Storage	Storage	Storage	Storage
Networking	Networking	Networking	Networking

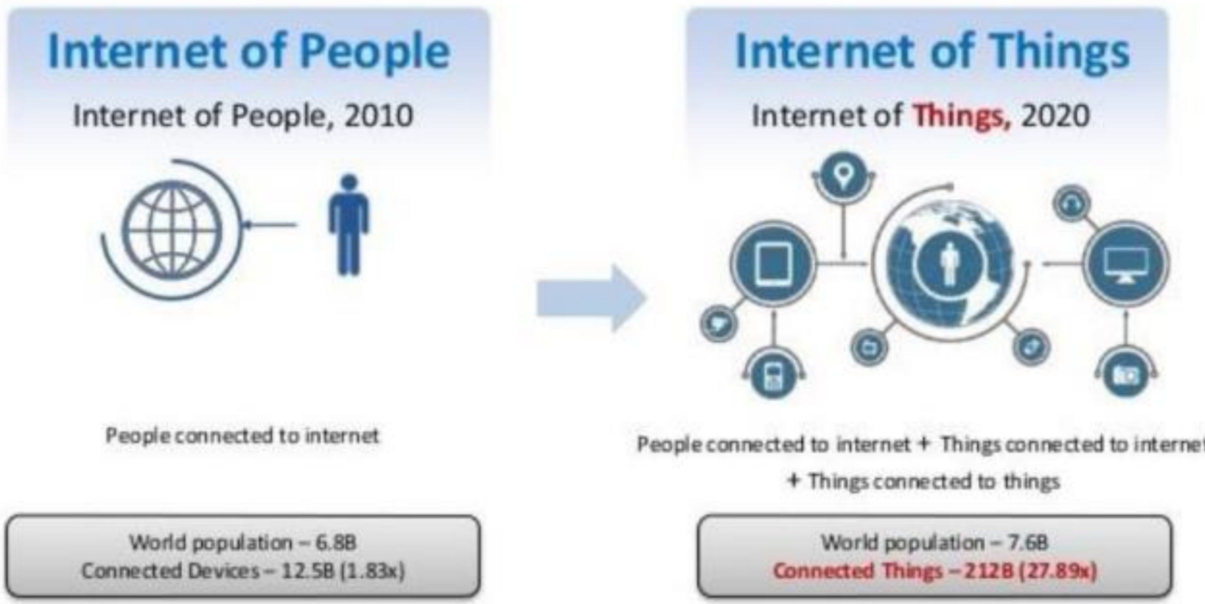
Managed by you

Managed by Vendor

INTERNET OF THINGS

Internet of Things (IoT)

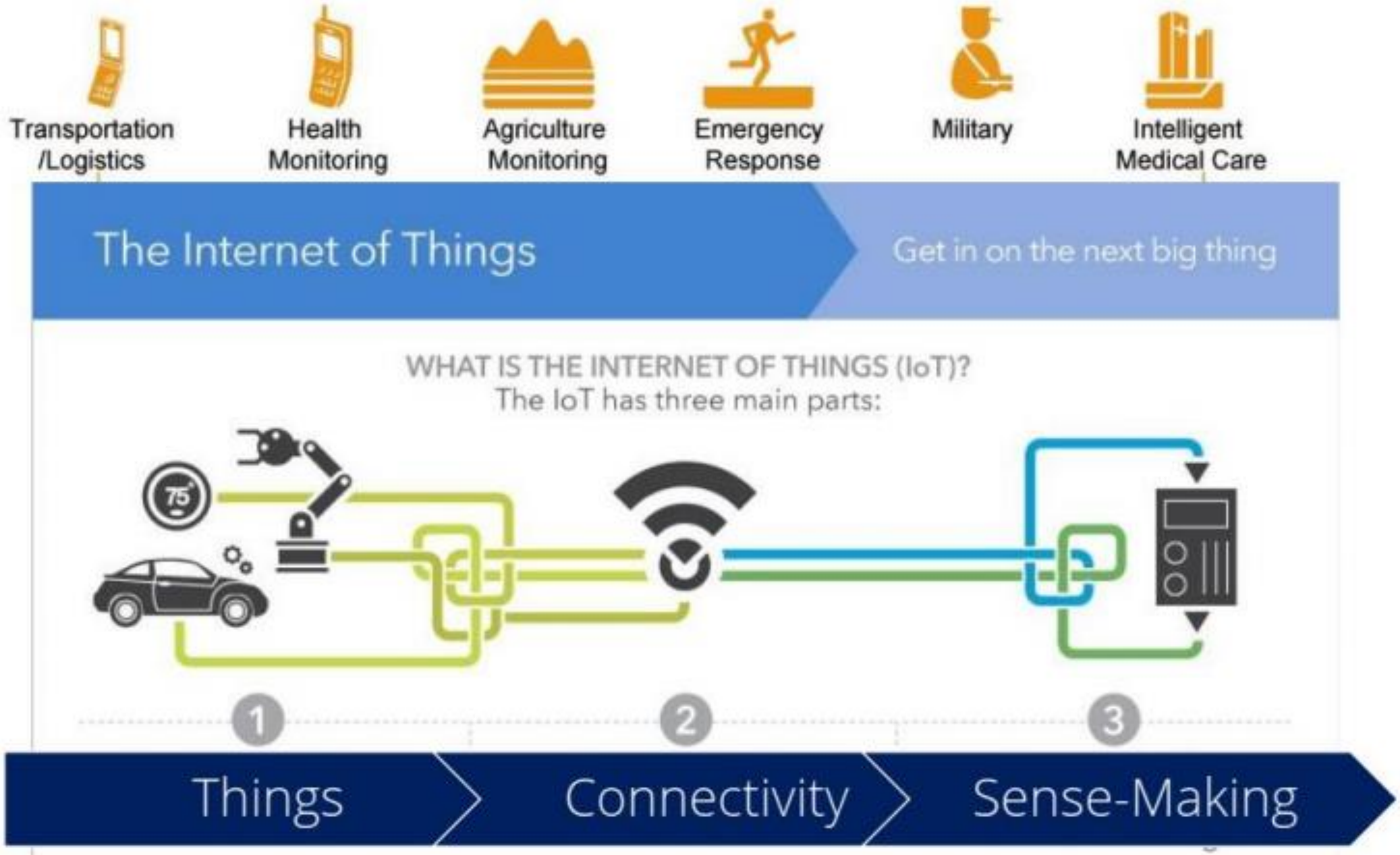
- A network of physical objects or "Things" using embedded electronics
- It enables "Things" to exchange data – machine to machine communication



From dumb to smart things
- Sense 'contextual info' in the background
- Share with other 'things' across internet
- Make decisions / take action autonomously and anticipatively

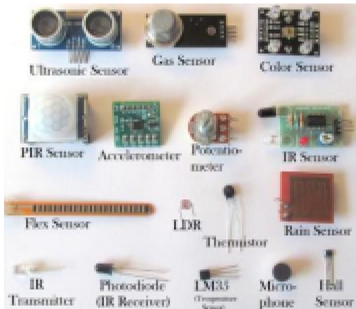
Leaving people time to pursue the more important things in life!

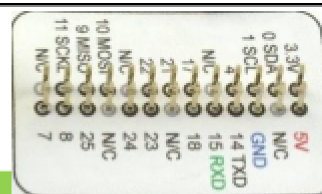
Internet of Things (IoT)

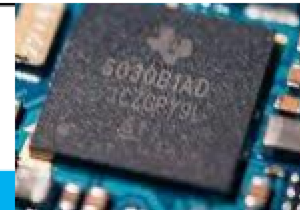


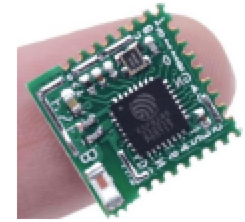
Sources: Intel and ZTE, https://www.linkedin.com/pulse/internet-things-iot-dummies-rajat-kochhar

IoT Smart Objects









Sensors

Actuators

I/O

CPU

Memory

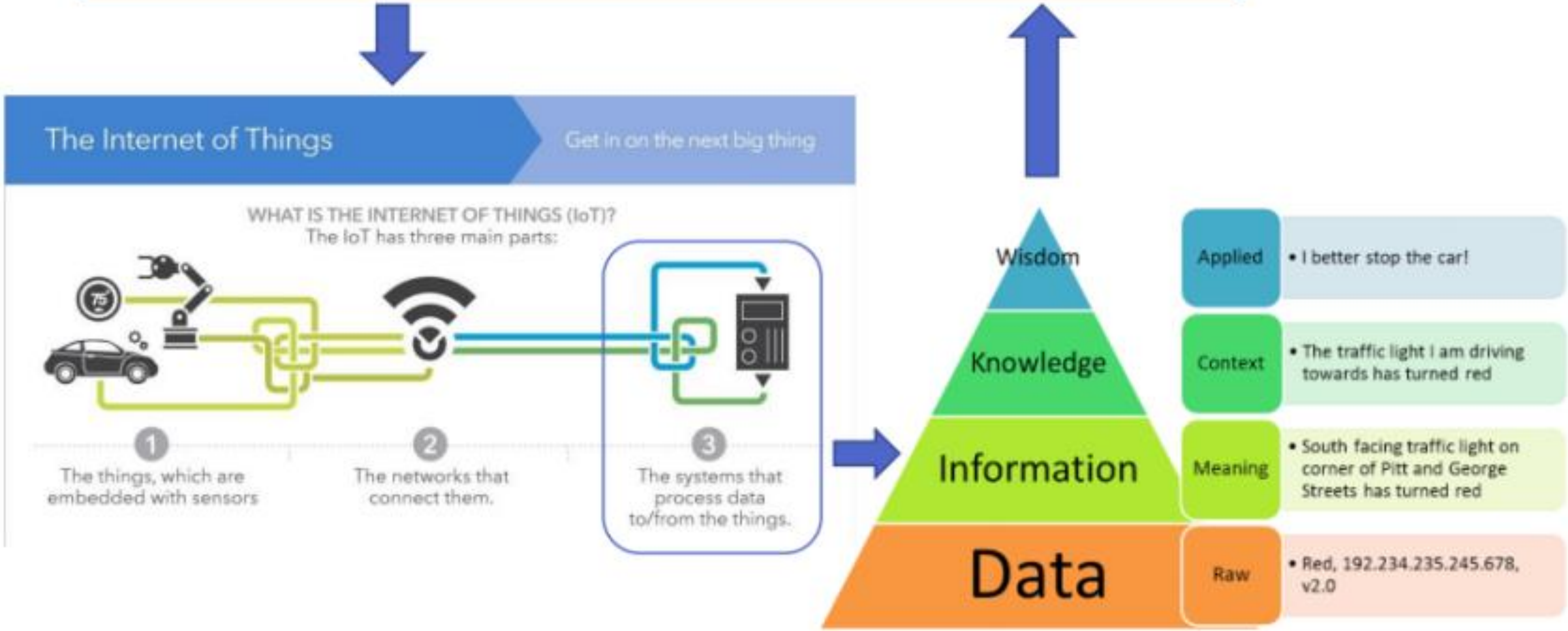
Net Work module

- 868 MHz
- 915 MHz
- 2.4 GHz
- 5 GHz

Power Sources

How can IOT Solve Problems

Problem Statement: How can IoT help stakeholder X decide/plan/take action to solve problem Y?



- Sources: Intel and ZTE, https://www.linkedin.com/pulse/internet-things-iot-dummies-rajat-kochhar

Role of Things

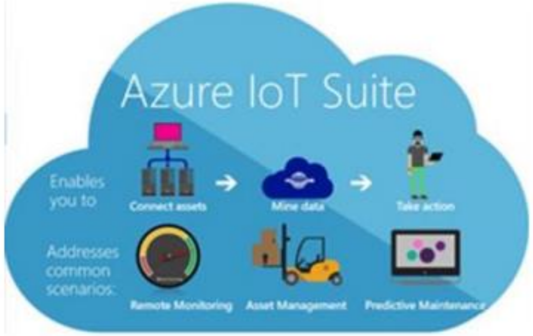
- Mimic the role of people
  - Ability to express context = sensors
  - Ability to respond = actuators
  - Intelligent = embedded processing/memory
  - Energized = mains/battery/energy-harvesting
  - Identifiable = unique addressing
  - Locatable = positioning
  - Reachable = connectivity (wired/wireless)

Role of Connectivity

- Flat connectivity (Things – Sense-making infrastructure)
  - Direct connectivity to the Internet without deploying addition infrastructure
  - Typically bi-directional (e.g., cellular, existing WiFi, Ethernet)
- Tiered connectivity (Things – Gateway – Sense-making infrastructure)
  - Gateway – translator for legacy things and sense-making infrastructure
  - Secure and reliable communication
    - Thing to Gateway: WiFi, IEEE 802.15.4, etc.
    - Gateway to Internet: Cellular, Ethernet

Role of Sense-making

- Transform raw data into actionable wisdom
- Convert data to Information to Knowledge to Wisdom
  Example of action triggered
- Actuation
- Decision making and planning
- New application and service
- Real-time vs non real-time



MOBILE DEVELOPMENT TECHNOLOGIES

Mobile Development Technologies

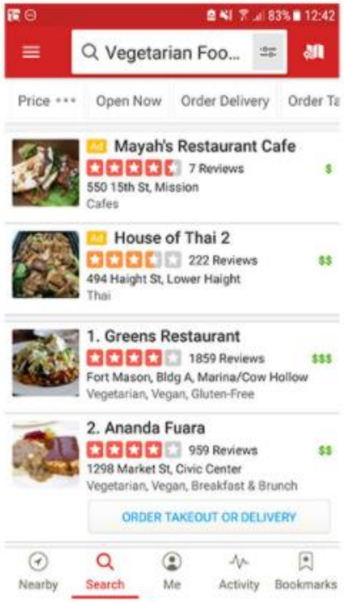
- Software used in mobile usually called “App”
- Mobile app is developed for mobile devices such as mobile phone, tablet, etc.
- Apps are generally downloaded from application platforms which are operated by the owner of the mobile operating system
  - App Store (iOS)
  - Google Play Store (Android)

Type of Mobile Apps

- Native Apps – coded in a specific programming language, such as Objective C for iOS or Java for Android.
    - Boast fast performance
    - High degree of reliability
    - Designed only for one type of operating system

Type of Mobile Apps

- Mobile Web Apps – actually website wich are designed in such a way that they look and feel like a mobile app
  - Run on smartphone browser, typically coded in HTML5



Mobile App

Web App

Type of Mobile Apps

- Hybrid Mobile Apps – part native and part web app
  - Like native apps, they can be found in app store and developed Native platform with the help of special container (Ionic, Sencha, PhoneGap)
  - Like web apps, they commonly rely on HTML5 and JavaScript

Mobile Development Technologies

- Top programming language for mobile development
  - Java (Android)
  - Python
  - HTML5
  - PhP
  - Swift (iOS)
  - Objective C (iOS)

Mobile Development Technologies

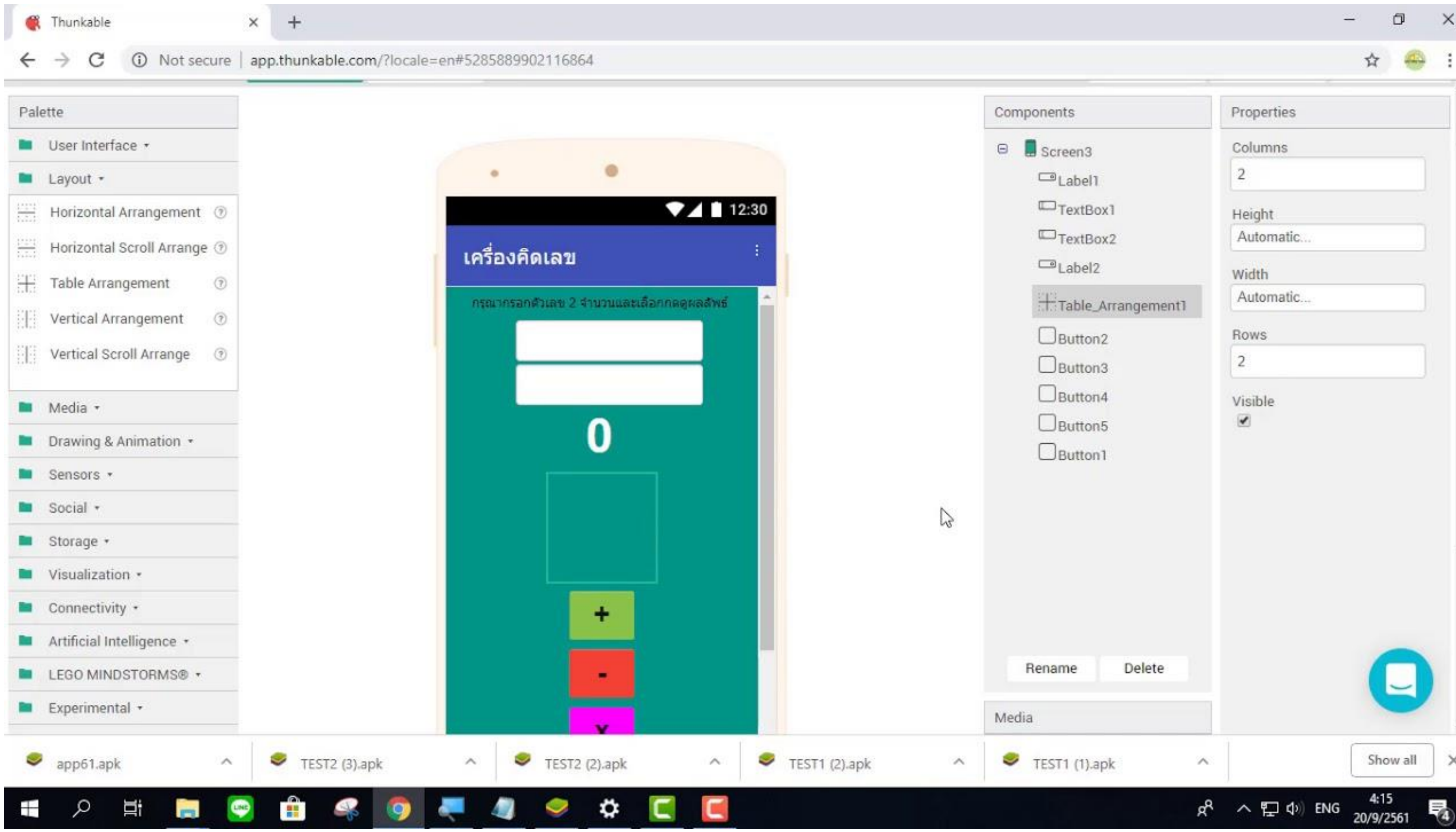
- Top framework or SDK for mobile development
  - Flutter by Google (cross-platform)
  - React Native (cross-platform)
  - Kotlin (cross-platform but prefer Android) – Java based
  - Ionic (iOS) – HTML5, JS, CSS based

Mobile Development Technologies

- Top tools/IDE for mobile development
  - Thunkable
  - Android studio
  - Xcode

Mobile Development Technologies

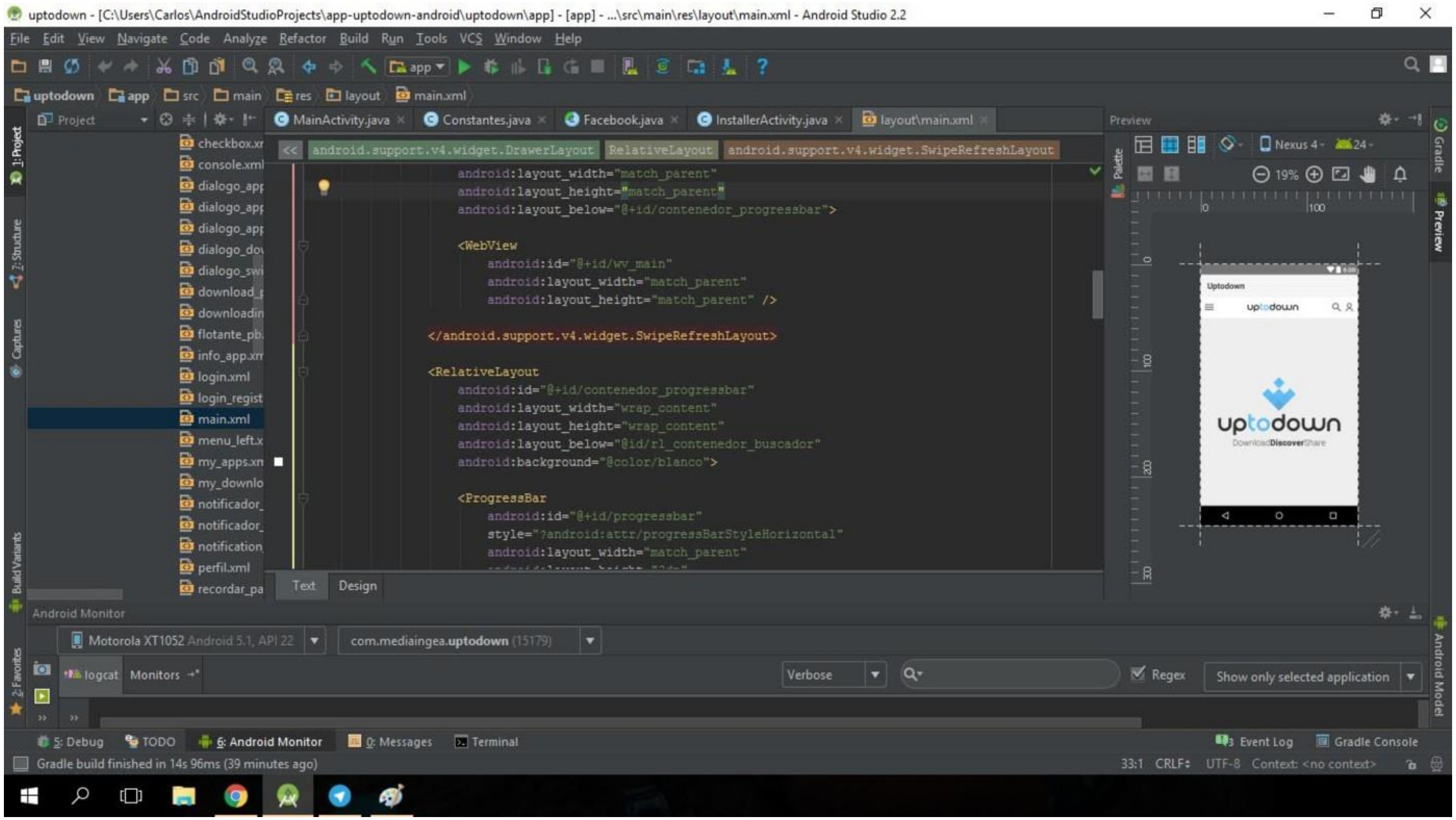
- Thunkable



Source: https://www.youtube.com/watch?v=cd_JUWTXqJE

Mobile Development Technologies

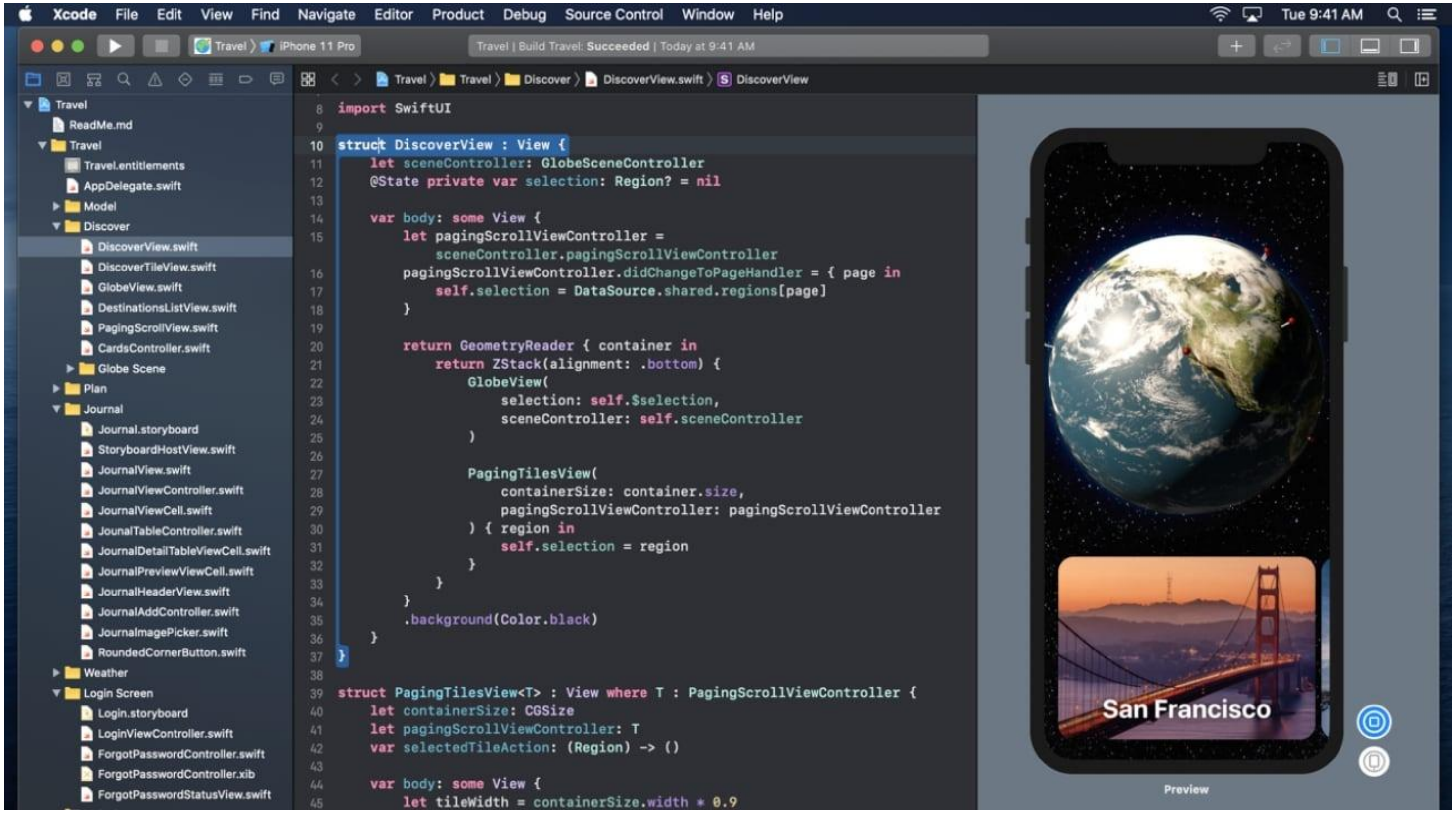
- Android Studio



Source: https://android-studio.th.uptodown.com/windows

Mobile Development Technologies

- Xcode



Source: https://appleinsider.com/articles/20/04/20/xcode-may-be-coming-to-the-iphone-and-ipad-very-soon